

东南大学考试卷 (A 卷)

课程名称 操作系统原理及应用 考试学期 17-18-3 得分
适用专业 软件工程 考试形式 闭卷 考试时间长度 120 分钟

题目	一	二	三	四	五	六	七	八	总分
得分									
批阅人									

1. Answer the following statement true or false by marking "√" or "x" on the examination paper. [20 marks]

- () (1) The operating system is responsible for resource allocation between competing processes.
- () (2) Disk latency mainly comprise rotational latency and transfer rate latency.
- () (3) Opening a file does not change to privilege mode of the processor.
- () (4) If the state of a process transfers from "waiting" to "ready", this implies that one interrupt has been generated in OS.
- () (5) Each thread has its separate set of register values and a separate stack.
- () (6) The context switch overhead of a user-level thread is more than the overhead for processes.
- () (7) When a process is in the critical section, it must be running.
- () (8) One solution to the "dining philosophers" is to always let hungry philosophers first grab the chopstick on his left hand and then that on his right hand. This solution can result in starvation.
- () (9) If x is a condition variable, $x.\text{signal}()$ always changes the state of x .
- () (10) The LRU and OPT page replacement algorithms never suffers from Belady's Anomaly.
- () (11) Deadlock prevention provides a set of methods for ensuring that at least one of the necessary conditions cannot hold.
- () (12) Binding of instructions and data to memory addresses can only happen at compile time or load time.

- () (13) The execution-time address binding scheme results in identical logical and physical addresses.
- () (14) Fixed-sized contiguous memory allocation suffers from internal fragmentation.
- () (15) With TLB, every data/instruction access requires only one memory access.
- () (16) When valid-invalid bit of page table is set to "invalid", the page either is not valid or is currently on disk.
- () (17) After a page-fault trap, it is necessary to restart the instruction which cause the trap.
- () (18) Local replacement algorithm can prevent the thrashing entirely.
- () (19) It is very important for an acyclic-graph directory structure to avoid cycles, so we can never create a hard link on a directory in such file systems.
- () (20) Polling, DMA and interrupt are three main interaction models between I/O controllers and CPUs to accomplish a complete I/O transfer.

2. Short Answer Questions [30 marks] (Please write the answers on the answer sheet.)

- a) Compare the concepts of thread and process. (6 marks)
- b) How to select the page size? Please give four considerations at least. (6 marks)
- c) In a time-sharing operating system, why is system performance so sensitive to the value that is selected for the time slice? Briefly explain what type of system behavior we would be likely to observe if the value selected for the time slice were too large? too small? (2+2+2=6 marks)
- d) Consider the following C program. Including the original parent process, how many processes are generated altogether? Show your reasons. (6 marks)

```
int main(int argc, char *argv[]) {
    int id0, id1, id2;
    id0 = fork ();
    id1 = fork ();
    id2 = fork ();
    if (id0 + id1 * id2 > 0) fork ();
    printf ("I am %d\n", getpid());
}
```

- e) A disk drive has 200 cylinders numbered 0 to 199. The read/write head of a disk drive currently lies at Cylinder 153, and the moving direction of the head is towards the Cylinder 0. Suppose the disk has pending requests, in FIFO order, trying to access the following cylinders: 86, 74, 109, 61, 98, 123, 150, 15, 172. Starting from the current head position, draw the graphs for (1) the SSTF, (2) the C-SCAN, and (3) the LOOK disk-scheduling algorithm. (2+2+2=6 marks)

3. Deadlock [10 marks] (Please write the answers on the answer sheet.)

A computer system has 3 types of resources A, B, and C with different numbers of instances. There are 4 running processes P1, P2, P3, and P4. The total resources, the resource's Allocation and Max matrixes for the four processes are shown as follows:

Process	Allocation			Max		
	A	B	C	A	B	C
P1	1	3	1	6	5	3
P2	0	2	2	3	5	3
P3	2	0	0	3	5	2
P4	0	1	3	2	4	3
Total resources	6	9	6			

- What are the 4 necessary conditions for a deadlock to occur? (2 marks)
- What are the matrixes Need and Available for the system? (2 marks)
- Please check if the system is currently deadlocked. Show your steps clearly. (2 marks)
- At the current state, if P2 requests additional resources [1,0,0], can the request be granted without any possible deadlock? Show your steps clearly. (2 marks)
- At the current state, if P1 requests additional resources [1,0,0], can the request be granted without any possible deadlock? Show your steps clearly. (2 marks)

4. Memory Management [10 marks] (Please write the answers on the answer sheet.)

The Intel IA-32 architecture was the dominant 32-bit industry chip for historical reasons. It supports segmentation and segmentation with paging and the common page size is 4KB. Each page table entry is 4-bytes in size, which consists of a physical frame number PFN (20 bits), a present bit P to indicate whether this page is in physical memory, a dirty bit D to indicate whether the page has been modified, an accessed bit A to track whether a page has been accessed, 6 bits for security or other uses, and 3 bits are ignored for reservation.

- What is the maximum size of physical memory space does this vanilla architecture can support? Show your reasons. (1+2=3 marks)

Assume there is a process with 3 virtual pages and 2 assigned physical frames. Its

first virtual page (VPN=0) resides in frame address 00101H, its second virtual page is not currently in memory, its third virtual page resides in frame address 01254H.

- Finish the following page table. (3 marks)

PFN										ignored		D	A						
										/	/	/	/	0	1	/	/	/	/
										/	/	/	/	0	0	/	/	/	/
										/	/	/	/	0	1	/	/	/	/

Suppose TLB is initially empty. Further suppose, the access time of each memory address be 100ns, the access time of TLB can be ignored, and the average time cost of handling a page fault (including the time of updating TLB and page table) is about 10⁷ns, use LRU with local replacement.

- Compute the effective access time for the following list of accessed virtual address: 2632H, 1368H, 2CA5H. Show your steps clearly. DO NOT rounded your results and you need not do the arithmetic to get full credit. (4 marks)

5. Page Replacement [10 marks] (Please write the answers on the answer sheet.)

- What is the LRU (least recently used) page replacement algorithm? (2 marks)
- Using the page references string shown below, fill in the frames and missing information to show how the LRU page replacement algorithm would operate. Use a dash "-" to fill in blank locations. Note that when there is more than one page that is a possible victim, always choose the one with the lowest frame number. (3 marks)

Num	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Refs	A	B	C	D	B	E	C	G	D	A	G	D	B	E	C
Frame1															
Frame2															
Frame3															
Frame4															
Fault?															

- What is the OPT(optimal) page replacement algorithm? (2 marks)
- Using the page references string shown below, fill in the frames and missing information to show how the OPT page replacement algorithm would operate. Use a dash "-" to fill in blank locations. Note that when there is more than one page that is a possible victim, always choose the one with the lowest frame number. (3 marks)

Num	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Refs	A	B	C	D	B	E	C	G	D	A	G	D	B	E	C
Frame1															
Frame2															
Frame3															
Frame4															

[illegible]

6. File System [10 marks] (Please write the answers on the answer sheet.)

Imagine a file system which uses inodes to manage files on disk. Each inode consists of a file name (4 bytes), user id (2 bytes), three timestamps (4 bytes each), protection bits (2 bytes), a reference count (2 bytes), a file type (2 bytes), and the file size (4 bytes). Additionally, the inode contains 13 direct indices, 1 index to a single indirect block, 1 index to a double indirect block, and one index to a triple indirect block. Each of these indices (block pointer) is 4 bytes. The file system also stores the first 356 bytes of each file in the inode.

- Three major methods of allocating disk space are introduced in our textbook. What are these three allocation methods? Which one is used in the previous file system? (3+1+4 marks)
- Assume a disk sector is 512 bytes and that each indirect block fills a single sector. What is the maximum file size for this file system? Show your work clearly. You need not do the arithmetic to get full credit. (5 marks)
- Is there any benefit to including the first 356 bytes of the file in the inode? If so, what is the reason? If not, why not? (1 mark)

7. Synchronization [10 marks] (Please write the answers on the answer sheet.)

Suppose there are two workers A and B in a factory. They take components from the same red box and assemble them into machines respectively. Then they put the assembled machines into another blue box. Make sure that they take components from the red box mutual exclusively, and put machines into the blue box alternately (in the order: A, B, A, B ...). Write C-like pseudo code that performs the appropriate initializations and enforces this execution order. Use only semaphores for your synchronization.

